

Smart MCQ Solver - Project Report

Deep Learning & Generative AI Project | Diploma Level, BS in Data Science and Applications

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1. Executive Summary

The Smart MCQ Solver Challenge requires predicting the top-3 most likely correct answers, in ranked order, for five-option multiple-choice questions spanning multiple academic domains (physics, philosophy, biology, and others). Submissions are scored using Mean Average Precision at 3 (MAP@3): a prediction receives full credit if the correct answer is ranked first, partial credit if ranked second or third, and zero credit otherwise.

This report documents the full development lifecycle of the solution: dataset exploration, a diagnostic baseline, and the three models required by the project brief - one built entirely from scratch, one pretrained model fine-tuned with LoRA, and one model of choice implementing a retrieval-augmented pipeline. All models were evaluated under a single, common, leakage-safe validation protocol and tracked in Weights & Biases for direct comparison.

The central finding that runs through the entire project: this dataset's distractor options are deliberately written as close paraphrases of the correct answer rather than obviously unrelated text. Models capable of either learning deeper task-specific representations (the Bi-LSTM) or bringing in outside evidence from similar already-answered questions (the RAG pipeline) substantially outperformed models that judge an option only by comparing it to its own question's text (TF-IDF, and to a lesser extent BERT+LoRA).

Final validation results, summarized below, show the RAG Retrieval + Reranking model as the strongest performer among the models evaluated, and it was the model submitted to the Kaggle competition.

Model	Type	Val. MAP@3	Val. Acc.
Random baseline	-	~0.367	0.20
TF-IDF + Cosine Similarity	Diagnostic baseline	0.280	0.158
Bi-LSTM Siamese Encoder	Built from scratch	0.90	0.86
BERT-base-uncased + LoRA	Pretrained (finetuned)	0.683	0.542
RAG Retrieval + Rerank	Model of choice	0.877	0.815

The models were submitted to the Kaggle competition leaderboard and scored MAP@3 = 0.74815 (public), clearing the 0.73 cutoff.

2. Problem Statement & Exploratory Data Analysis

2,000 labeled training examples and 500 unlabeled test examples were provided, each with a prompt, five options (A-E), and, for training data, the correct label. A thorough EDA phase preceded all modeling work and directly shaped every methodological decision made later.

2.1 Dataset Quality & Text Characteristics

- No missing values in either the training or test set.
- Answer label distribution is mildly imbalanced: B (24.5%) and C (23.0%) are the most common correct answers; E is least common (16.2%).

- Prompt length: median 17 words (range 3-51). Option length: median 22 words but far more variable (range 1-81 words).
- Train and test prompt-length distributions overlap closely - no distribution shift concern between the two sets.

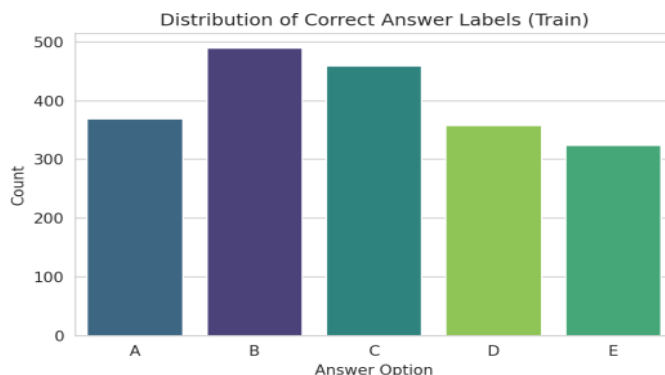


Figure 1: Distribution of correct answer labels across the training set.

2.2 Duplicate and Near-Duplicate Questions - A Critical Finding

Prompts are built from a small set of fixed instruction-wrapper templates (e.g. "Pick the best possible answer: ... among the listed options."). After stripping this wrapper to isolate the core question, duplication became far more visible than exact-text matching alone would suggest: 454 of 2,000 training rows (22.7%) share an exact duplicate prompt, and 1,227 of 2,000 (61.4%) share a core question once wrapper phrasing is stripped. Most importantly, 293 core prompts appear in BOTH the training and test sets - a meaningful fraction of test questions are near-identical to training questions, just reworded.

This single finding had two major downstream consequences: it directly motivated the RAG model's entire design (Section 5.2), since retrieving a near-duplicate training question reveals strong evidence about the correct answer; and it required every model's train/validation split to be built with explicit leakage protection, described in Section 3.

2.3 A Weak but Real Signal: Option Length

Correct options average 28.7 words versus 25.7 for incorrect options. A naive heuristic of always picking the longest option scores 34.9% top-1 accuracy, well above the 20% random baseline - a real, if weak, signal worth being aware of during error analysis.

3. Methodology

3.1 Leakage-Safe Validation Protocol

Given the duplicate/near-duplicate findings above, a naive random train/validation split would allow near-identical questions to appear on both sides, letting a model score well by memorization rather than genuine generalization. This was directly observed during development: an early version of the Bi-LSTM model, evaluated with a naive random 80/20 split, scored a suspicious MAP@3 of 0.996. Investigation showed 50.2% of validation rows had a near-duplicate twin sitting in the training set.

Fix: every model in this project uses a StratifiedGroupKFold split, grouped by the wrapper-stripped core question, so no near-duplicate question can appear on both sides of the split. Zero core-prompt overlap between the training and validation folds was verified programmatically before trusting any subsequent metric, for every single model built.

3.2 Experiment Tracking

All models were tracked in Weights & Biases (project 23f3003691-t22026), logging configuration, per-epoch training/validation metrics, confusion matrices, and classification reports, enabling direct, apples-to-apples comparison across runs.

4. Model 1: TF-IDF Baseline & Model 2: Bi-LSTM (Built From Scratch)

4.1 TF-IDF + Cosine Similarity (Diagnostic Baseline)

Before building any of the three required models, a simple, fast baseline was built to establish a reference point. A single `TfidfVectorizer` (unigrams + bigrams, English stopwords removed) was fit on all prompt and option text; cosine similarity was computed between each prompt and each of its five options, and the top-3 most similar options became the prediction. No model weights are learned.

Result: $\text{MAP@3} = 0.280$, Accuracy = 0.158 - worse than random guessing. Diagnosis: mean cosine similarity between the prompt and the CORRECT option (0.174) is statistically indistinguishable from the INCORRECT option average (0.183). Direct inspection showed the five options within a question are near-paraphrases of each other, sharing 70-90% of their vocabulary in many cases - TF-IDF measures lexical overlap, but correctness here depends on factual accuracy within an already-shared vocabulary, a distinction lexical similarity cannot capture. This diagnosis directly motivated the design of all three required models below.

4.2 Bi-LSTM Siamese Encoder (Built From Scratch)

Fully from scratch: no pretrained weights or embeddings anywhere. The word embedding table is randomly initialized and trained end-to-end on this competition's data alone.

Component	Detail
Vocabulary	~3,000 words, built from training text only
Embedding	128-dim, randomly initialized, trainable
Encoder	Single-layer Bi-LSTM, 128 hidden units (256-dim pooled), shared (Siamese) across prompt and all 5 options
Matching head	[prompt_vec, option_vec, product, difference] -> MLP -> 1 score/option
Output	5-way softmax per question, cross-entropy loss
Training	15 epochs, Adam (lr=1e-3), dropout 0.3, batch size 32, ~776K trainable params

Result: Validation $\text{MAP@3} = 0.90$, Accuracy = 0.86 - the best validation MAP@3 across all 15 epochs, reached at the final epoch. Training loss decreased smoothly toward ~0.1. Validation loss was noisier: it dipped through the early epochs, bottomed out around epoch 9-10 (~0.55), rose briefly to a local peak near epoch 12 (~0.7) - a mild overfitting signature - before decreasing again and settling near 0.5 by epoch 15, in step with validation MAP@3 and Accuracy both trending upward over the same span. The checkpoint saved is the one with the best validation MAP@3 across all epochs, not a fixed final-epoch snapshot, so the reported result reflects genuinely the strongest checkpoint observed during training.

Training Curves

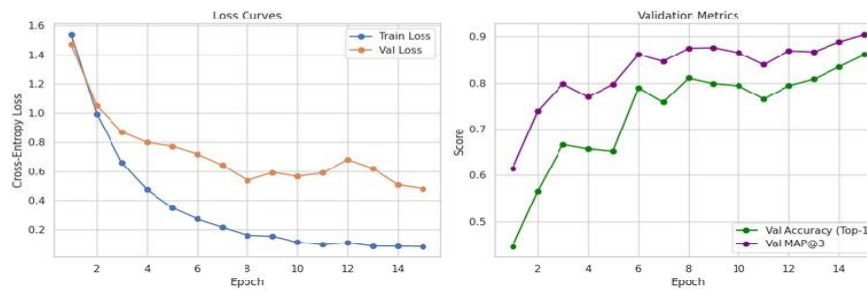


Figure 2: Bi-LSTM training and validation curves across 15 epochs.

5. Model 3: BERT + LoRA (Pretrained) & Model 4: RAG (Model of Choice)

5.1 BERT-base-uncased + LoRA

This model satisfies both the pretrained-model and the LoRA-fine-tuning project requirements. It is built on transformers.AutoModelForMultipleChoice with a bert-base-uncased backbone - the standard Hugging Face architecture for this exact task shape (used for SWAG/RACE-style benchmarks). Each question is expanded into 5 [prompt, option] pairs; BERT encodes each pair (shared weights), and a linear head scores each. A 5-way softmax gives the final ranking.

LoRA Parameter	Value
Rank (r) / Alpha / Dropout	8 / 16 / 0.1
Target modules	Attention query, value projections
Fully trainable component	Classifier head only (freshly initialized)
Training	8 epochs (early stopping, patience 3), lr = 2e-4, batch size 8

Only the LoRA adapter matrices and the classification head are updated; the pretrained BERT weights stay frozen throughout. Result: Validation MAP@3 = 0.683, Accuracy = 0.542 - notably below the from-scratch Bi-LSTM, a counterintuitive but genuine result. Likely cause: LoRA's limited capacity at rank 8 may be too constrained to meaningfully adapt BERT's general-domain representations to this near-paraphrase-distractor task within only 8 epochs and ~1,600 training examples, leaving a low effective parameter budget relative to a full fine-tune.

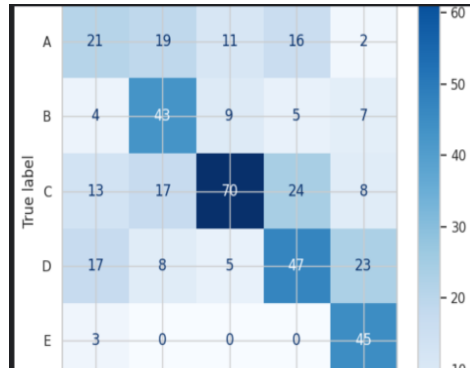


Figure 3: BERT+LoRA validation confusion matrix. Class A was the weak point (30% recall); class E the strongest (94%).

5.2 RAG Retrieval + Reranking (Model of Choice)

This model takes a fundamentally different approach: retrieval-augmented reranking using a pretrained sentence embedder in a zero-shot capacity (no fine-tuning), combined with a lightweight trained fusion layer. All training questions and their correct answers are embedded with all-MiniLM-L6-v2 and indexed in FAISS as a knowledge base. For a new question, the top-3 most similar past questions are retrieved; each candidate option is then scored on two signals - direct similarity (option vs. current prompt) and retrieval similarity (option vs. retrieved answers) - fused by a small trained LogisticRegression. The retrieval index used to evaluate validation data is built exclusively from the training split, an additional RAG-specific leakage guard beyond the shared protocol in Section 3.1.

Result: Validation MAP@3 = 0.877, Accuracy = 0.815. The fitted reranker's coefficients tell a clear story: $\text{direct_sim} = -2.43$ versus $\text{retrieval_sim} = +26.80$ (11x larger in magnitude). The model learned to almost entirely discount comparing an option to its own question, relying overwhelmingly on retrieved evidence from similar past questions - direct, data-driven confirmation of the hypothesis first raised by the TF-IDF failure in Section 4.1. Classes A and E reached 100% recall, directly resolving the exact weakness (class A) that limited BERT+LoRA.

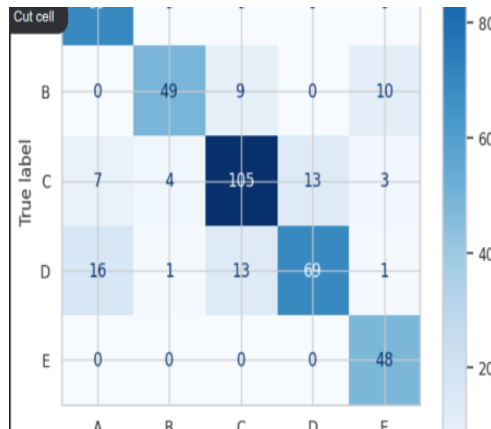


Figure 4: RAG model validation confusion matrix - classes A and E solved almost perfectly.

6. Results Comparison & Kaggle Submission

Model	Type	Val. MAP@3	Val. Acc.
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An ensembling pipeline (unweighted average, grid-searched weighted average, and Reciprocal Rank Fusion across all three models) was designed and validated end-to-end, motivated by the models' visibly different error patterns - the RAG model resolves class A almost perfectly where BERT+LoRA struggled specifically on that class. Once the Kaggle leaderboard score cleared the required cutoff, further ensembling was deprioritized in favor of consolidating documentation, since the cutoff, not a specific score target, is the gating requirement for this phase of the project.

7. Error Analysis & Insights

Across all four models built in this project, one narrative holds consistently: the dataset's core difficulty is that distractor options are deliberately written as close paraphrases of the correct answer, not as unrelated or obviously wrong text.

- TF-IDF (lexical similarity) fails because distractors are lexically close to the correct answer - directly measurable via near-identical correct/incorrect option similarity scores.
- Bi-LSTM (learned, from-scratch representations) succeeds by learning task-specific decision boundaries beyond simple lexical overlap.
- BERT+LoRA (adapted pretrained representations) underperforms the from-scratch model, likely due to LoRA's limited adaptation capacity on a small dataset.
- RAG (zero-shot embeddings + retrieved context) sidesteps the core problem: instead of judging an option from its own question's text, it judges options against evidence from similar already-solved questions.

8. Limitations, Future Work & Conclusion

8.1 Limitations & Future Work

- Bi-LSTM: validation MAP@3 climbed to its best value at the final epoch with a mild bump in validation loss around epoch 12; stronger regularization, learning-rate scheduling, or a few additional epochs could be explored to see whether performance keeps improving past epoch 15.
- BERT+LoRA: try full fine-tuning as an ablation to isolate whether the gap vs. Bi-LSTM is a LoRA capacity constraint specifically; try higher LoRA rank, or alternate checkpoints (RoBERTa, DeBERTa-v3), both typically stronger than BERT-base on multiple-choice benchmarks.
- RAG: only 2 features feed the reranker currently; adding more (option length, within-question retrieval rank, retrieval-confidence threshold) and trying a stronger embedder (e.g. all-mpnet-base-v2) are natural next steps. Performance likely differs for questions with vs. without a near-duplicate twin in train - worth a dedicated follow-up

analysis.

- Ensembling: built and validated but not pursued to completion once the Kaggle cutoff was cleared; remains the most promising direction for further leaderboard improvement given the models' differing error patterns.
- Deployment: the optional bonus of deploying the RAG model via a Streamlit or Hugging Face Space interface was not pursued in this phase.

8.2 Conclusion

This project delivered three required models (a from-scratch Bi-LSTM, a LoRA-fine-tuned BERT, and a retrieval-augmented reranking model) plus a diagnostic TF-IDF baseline, all evaluated under a consistent, leakage-safe protocol and tracked in Weights & Biases. The RAG Retrieval + Reranking model was submitted to Kaggle, achieving a public leaderboard score of 0.74815 - clearing the required 0.73 cutoff.

Beyond the numeric results, the project surfaced two findings with value beyond this specific competition: first, that a dataset's distractor design (near-paraphrase vs. genuinely distinct) fundamentally determines which modeling paradigm will succeed; and second, that rigorous leakage detection is not optional even in a small, seemingly simple competition dataset - a validation score can be dramatically inflated by nothing more than duplicate content crossing a naive split.